

Minimization and Maximization Problems: When Does Setting the Gradient to Zero Work?

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Every MLE note in this repository solves $\nabla(\cdot) = 0$ at some point. This page states precisely when that move is justified, what can go wrong, and what replaces it under constraints (Lagrange, KKT). `optimization_demo.py` demonstrates the failure modes and checks the constrained solutions numerically.

1 Unconstrained problems: stationarity

Maximization is minimization of the negation, so we speak of $\min_{\theta} f(\theta)$ throughout.

Theorem 1 (First-order necessary condition, Fermat). *If θ^* is a local minimizer of differentiable f and lies in the interior of the domain, then*

$$\nabla f(\theta^*) = 0. \tag{1}$$

Necessary — not sufficient. The catalogue of failures:

- **Saddles and maxima.** $f(x, y) = x^2 - y^2$ has $\nabla f = 0$ at the origin, which is neither (script, panel 1). Second-order check: $\nabla^2 f \succ 0$ at a stationary point $\Rightarrow$ local min; indefinite $\Rightarrow$ saddle.
- **Boundary optima.** On a constrained domain the minimum can sit on the boundary with $\nabla f \neq 0$ (Section 3 handles this properly).
- **Non-differentiability.** $f(x) = |x|$ minimizes at 0 with no derivative there; subgradients ($0 \in \partial f$) generalize (1).
- **Solvable $\neq$ solvable in closed form.** For logistic regression, $\nabla = 0$ has no algebraic solution — the equation is transcendental. That is not a failure of stationarity but of algebra; it is *why iterative methods exist* (cf. the optimizers page). Our Bernoulli/Gaussian MLEs are the lucky cases where the stationary equations untangle.

2 Convexity: when stationarity is the whole story

Definition 1. f is convex if $f(\lambda u + (1 - \lambda)v) \leq \lambda f(u) + (1 - \lambda)f(v)$ for all u, v and $\lambda \in [0, 1]$. Differentiable characterizations: f convex $\iff f(v) \geq f(u) + \nabla f(u)^\top (v - u)$ (tangents underestimate) $\iff \nabla^2 f \succeq 0$.

Theorem 2. *For convex differentiable f , every stationary point is a global minimizer; if f is strictly convex, the minimizer is unique.*

This theorem is the license behind every “set the derivative to zero” in this repository: the Bernoulli log-likelihood (strictly concave), the Gaussian NLL in each coordinate, least squares with full-rank X . Two quantitative grades refine convexity, and between them govern first-order convergence rates:

$$\underbrace{f(v) \geq f(u) + \nabla f(u)^\top (v - u) + \frac{\mu}{2} \|v - u\|^2}_{\mu\text{-strong convexity}}, \quad \underbrace{\|\nabla f(u) - \nabla f(v)\| \leq L \|u - v\|}_{L\text{-smoothness}}, \quad (2)$$

with condition number $\kappa = L/\mu$: GD contracts the error by $(1 - 1/\kappa)$ per step — geometry, not luck, sets the speed. *Convex on a bounded domain* adds existence: a continuous convex f on a compact convex set attains its minimum (Weierstrass), even when stationarity never fires because the optimum is on the boundary.

3 Constraints I: equalities and Lagrange multipliers

Problem: $\min f(\theta)$ s.t. $h(\theta) = 0$. At a constrained optimum nothing can be gained by moving *along the constraint surface*, so ∇f must be normal to it:

$$\nabla f(\theta^*) + \lambda \nabla h(\theta^*) = 0, \quad h(\theta^*) = 0, \quad (3)$$

the stationarity of the *Lagrangian* $\mathcal{L}(\theta, \lambda) = f(\theta) + \lambda h(\theta)$ in both arguments. The multiplier is a price: $\lambda = -\partial f^*/\partial c$ for the perturbed constraint $h = c$. Worked instance (script, panel 2): $\min x^2 + y^2$ s.t. $x + y = 1$ gives $(x, y) = (\frac{1}{2}, \frac{1}{2})$, $\lambda = -1$ — and the multinoulli MLE $\hat{p}_j = n_j/n$ of the distributions page is exactly this machinery applied to $\sum_j p_j = 1$.

4 Constraints II: inequalities and KKT

Problem: $\min f(\theta)$ s.t. $g_i(\theta) \leq 0$, $h_j(\theta) = 0$.

Theorem 3 (Karush–Kuhn–Tucker conditions). *Under constraint qualification (e.g. Slater: some strictly feasible point exists, for convex problems), a constrained local optimum θ^* admits multipliers μ_i, λ_j with*

$$\begin{aligned} \text{stationarity:} & \quad \nabla f + \sum_i \mu_i \nabla g_i + \sum_j \lambda_j \nabla h_j = 0 \\ \text{primal feasibility:} & \quad g_i(\theta^*) \leq 0, \quad h_j(\theta^*) = 0 \\ \text{dual feasibility:} & \quad \mu_i \geq 0 \\ \text{complementary slackness:} & \quad \mu_i g_i(\theta^*) = 0 \quad \text{for all } i. \end{aligned} \quad (4)$$

For convex f, g_i and affine h_j , the KKT conditions are necessary and sufficient for global optimality.

Complementary slackness is the operative line: each inequality is either *inactive* ($g_i < 0$, then $\mu_i = 0$ and it might as well not exist) or *active* ($g_i = 0$, and $\mu_i \geq 0$ prices it). Solving a KKT system means guessing the active set, solving the resulting equality problem, and checking signs — the script’s panel 3 walks the one-dimensional instance $\min(x - 2)^2$ s.t. $x \leq 1$: the

unconstrained answer $x = 2$ is infeasible, the constraint activates, $x^* = 1$, $\mu = 2 > 0$. KKT is the grammar under SVMs (support vectors are the active constraints), projected methods, and every ‘s.t.’ in an ML paper; Lagrangian *duality* — maximize $\min_{\theta} \mathcal{L}$ over the multipliers — is its global companion, with strong duality under Slater.